

AIRPORTS



TODAY, MORE PEOPLE TRAVEL by air than ever before. Whether they are business people off to visit clients or families going on holiday, all air travellers leave from airports, which range in size from small local facilities to enormous international terminals. A large airport is like a city. It contains shops, offices, and hotels, in addition to all the buildings, runways, and taxiways needed to service the aircraft and their passengers. Airport security is always tight, because airports and aircraft have often been the targets of terrorist attacks.

Features of an airport

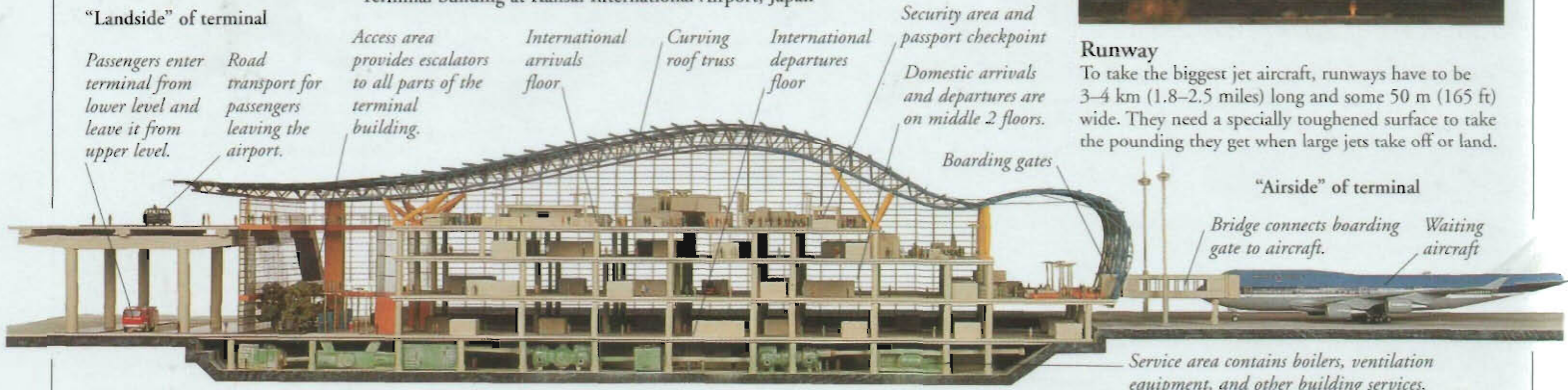
Aircraft take off and land on runways, which are linked to the terminal buildings by routes called taxiways. The passengers embark and disembark at the terminal buildings. For the aircraft, the airport has repair workshops, refuelling facilities, and storage hangars.



Runway

To take the biggest jet aircraft, runways have to be 3–4 km (1.8–2.5 miles) long and some 50 m (165 ft) wide. They need a specially toughened surface to take the pounding they get when large jets take off or land.

Terminal building at Kansai International Airport, Japan



Service area contains boilers, ventilation equipment, and other building services.

Air traffic control

At the heart of an airport is the control tower, where air traffic controllers monitor every moment of an aircraft's arrival and departure. They make sure that each pilot follows the correct flight path, that all aircraft land in the right place, and that there is a safe amount of time between each take-off and landing.



Air traffic controllers in the control tower

Radar display screen

Airport radar tracks each aircraft as it lands, giving the controllers precise details of its position. All aircraft within 20 to 50 km (12 to 30 miles) of the airport can be tracked by radar and shown on the controllers' display screens.

Flight path

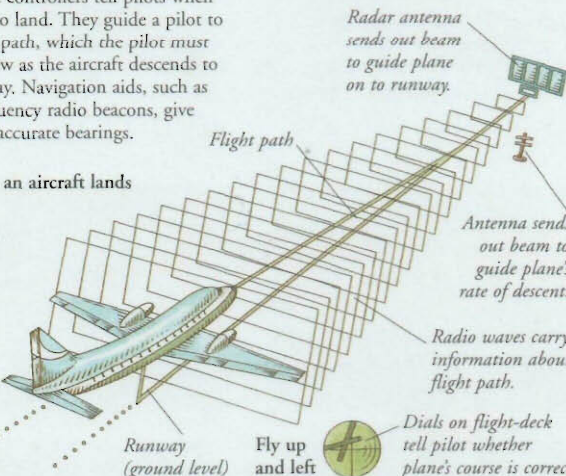
Air traffic controllers tell pilots when it is safe to land. They guide a pilot to a specific path, which the pilot must then follow as the aircraft descends to the runway. Navigation aids, such as high-frequency radio beacons, give the pilot accurate bearings.

How an aircraft lands

Fly down and right



Course is correct



Security

Airport security staff are always on their guard, trying to spot terrorists or smugglers. Metal detectors and other electronic devices alert staff when a passenger is carrying a gun or other type of weapon. There are also "sniffer" dogs that have been trained to detect the scent of explosives or illegal drugs.

Passports

A person travelling from one country to another usually carries a passport, an official document that identifies the owner and their place of origin. Passports are inspected at international airports.



EU passport

An X-ray reveals a gun.



X-ray scanner

Airport staff use X-ray machines to scan the contents of passengers' luggage. A screen on the side of the X-ray machine shows what is inside each bag. Different materials show up in different colours, enabling items such as guns to be found with ease.

Airports and the environment

A large airport can have a devastating impact on the local environment. Clearing the land to build an airport destroys carefully balanced ecosystems, while the air pollution can harm wildlife, and the noise may scare some animals away.

Airport ecosystems

Since airports cover such vast areas, birds and animals can also move into these areas and establish new ecosystems, undisturbed by people.



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